

Human-Centric Irregular RIS-Assisted Multi-UAV Networks With Resource Allocation and Reflecting Design for Metaverse

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Abstract—Human-centric Metaverse services requires novel communication and networking technologies to achieve seamless connectivity for Metaverse users. Reconfigurable intelligent surface (RIS) in 5G and beyond networks can provide highly reliable communication connections, superior user quality of service (QoS), seamless user connections, and extensive signal coverage for Metaverse. Deploying RIS in unmanned aerial vehicle (UAV) networks for Metaverse can enormously improve the signal propagation environment and human-centric communication experiences. Considering the channel uncertainty of the air-ground cascade communication link in Metaverse, an RIS-aided multi-UAV cross-layer network system is proposed. Under the cross-tier interference limitation and the rate outage probability constraint, the system EE improved by maximizing the minimal energy efficiency (EE) of UAV units. Different from the existing RIS schemes, which suffer from the significant channel acquisition cost or power consumption, this paper first proposes a topology design scheme of irregular RIS, which Metaverse user only connects a few RIS elements to obtain high EE. Secondly, with the imperfect cascade channel state information (CSI) error model, the rate outage probability constraint is approximated by Bernstein type inequality to enhance the seamless human-centric connectivity service. Hence a low complexity scheme is invoked to co-design the power control parameter at the UAV transmitter and RIS reflecting phase. Finally, affluent simulation curves verify that the irregular RIS controller deployment combined with low power loss topology design and low-complexity phase shift design contributes to improve human-centric QoS for Metaverse service.

Index Terms—Human-centric, metaverse, RIS, UAV, EE, imperfect cascaded CSI, topology design, phase shift design.

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I. INTRODUCTION

DUE to the demand for ultra-immersive service in the Metaverse, ultra-low and uniform network latency is required to achieve super interactivity [1], [2], [5]. However, the current communication and network infrastructure is unable to meet these requirements, as any slight deviation can lead to interruptions in the data flow from the Metaverse services. New technological breakthroughs in 5G and beyond networks can provide the Metaverse with the ultimate high-capacity and reliable connectivity, fully unleashing the potential of Metaverse services [3]. As an innovative key technology in 5G and beyond networks, RIS can reduce energy consumption and improve transmission reliability in Metaverse communication networks, achieving sustainable and stable services for the Metaverse [1]. As a potential technology for enhancing stable signal interaction within the Metaverse, RIS has gained widespread attention, particularly with performance analysis [4], [6], phase shift design [7], [8], etc. RIS can improve wireless connectivity by reducing power consumption and the randomness of electromagnetic wave transmission, thus potentially improving network coverage and system EE [8]. In the RIS-assisted system, the transmission reliability can be greatly improved by establishing virtual chains from communication base stations (BSs) to Metaverse users.

A. Related Works and Motivation

The Metaverse relies heavily on omnipresent continuous sensing and immersive instantaneous interactive data transfer, which requires a significant amount of energy [3], [9]. Energy efficiency is crucial in the RIS-assisted Metaverse service because the power loss introduced by large-scale RIS component deployment cannot be ignored [10]. Besides, the emergence of diverse energy-intensive Metaverse services highlights that it is imperative to reduce greenhouse gas emissions resulting from communication technologies [11]. Abundant works have been done to formulate the EE optimization framework and design energy efficient resource allocation and RIS controller design schemes in different networks [12], [13]. In [14], a novel RIS-enabled coordination multipoint model was proposed. The BS clustering, subcarrier allocation, power control, RIS optimization were considered to attain

the maximum system EE. In paper [15], the authors studied the nontrivial tradeoff between spectral efficiency and EE in multi-user multiple-input multiple-output (MIMO) system. An iterative minimization method was proposed by using homotopy, accelerated projection gradient and optimization minimization method to design RIS phase. In [16], the aerial RIS layout and the unit phase were designed together to achieve the UAV-RIS network EE by fixing RIS on the aerial facility. In [17], the authors attempted to acquire the maximum EE of the RIS-enabled link by seeking valid solutions of active and passive beamforming. Resource allocation involving distributed RIS was proposed in [18]. With the premise of meeting the minimum QoS, unit modulus constraints and maximum transmission power, the RIS switching state, phase shift and transmit beamforming were jointly designed to maximize EE. In [19], the authors established an RIS-enabled MIMO uplink transmission network model that adopted only part of the channel information, and statistical state information between users and RIS. A joint design method based on user transmit covariance matrix and IRS unit phase matrix was adopted to maximize global system EE.

However, all the above studies only follow perfect CSI at the transmitter. Unfortunately, collect perfect CSI in RIS-enabled system is arduous as a whole because no radio frequency (RF) chain is employed [20], [21]. Currently, there are two ways to deal with the imperfect channel information in the RIS auxiliary system. The first way is to acquire CSI of BS-RIS and RIS-users link separately. It is necessary to install the active transmit RF chain at the RIS to collect the two independent RIS-aided links. Related studies such as paper [22] proposed to maximize traversal and capacity in the case of statistical CSI. In [23], under probabilistic outage constraints, a security and EE maximization problem was proposed under channel uncertainty. The authors of [24] studied a robust frame for RIS-enabled mmWave vehicle-borne communication, which maximized the uplink average rate by only relying on imperfect statistical CSI. Reference [26] proposed a novel MIMO symbiotic radio system based on uplink RIS to realize the weighted sum-rate maximization. The disadvantage of this approach was that it required additional power and circuit costs. The 2nd way was to consider the joint cascaded CSI of BSs-RIS-users link. This method had the advantage of estimating cascaded channels without additional hardware and without power cost. Therefore, the method of cascade channel estimation attracts more research. Follow the goal of minimizing transmit power, in [25], the beamforming, artificial noise and RIS phase shift matrices were co-designed based on the outage probability restriction under the cascaded CSI model. The work in [29] aimed to provide a framework for insight of the performance of RIS-enabled large-scale MIMO transmission with imperfect CSI. In paper [30], robust beam design under various CSI error models were explored separately for finding the minimum transmit power target.

Since the terrestrial network link signal is interfered by obstacles from time to time, we introduce UAVs into the RIS-aided communication system in Metaverse to improve signal connectivity [2]. As an air base station, the UAV can be flexibly deployed and has a higher chance of establishing

an air-ground line-of-sight (LoS) communication link, and therefore has an excellent wireless channel [31], [32], [33]. In Metaverse, the combination of RIS and UAV has many advantages [17]. RIS can help establish virtual communication links, and UAV can establish LoS communication links as much as possible. The combination of them has the double effect of enhancing the desired signal or suppressing interference. The complementary advantages of RIS and UAV can bring significant signal transmission performance improvements to Metaverse network, enhancing the experience of a seamlessly connected network for human-centered users [3], [18]. However, while the combination of UAV and RIS can improve the LoS propagation of Metaverse signals and blind signal coverage, this also introduces significant power consumption from UAVs. Likewise, the power loss associated with the large-scale deployment of RIS components cannot be ignored. Driven by these factors, we consider the EE maximization study of RIS-UAV Metaverse network services under imperfect cascade CSI. On the one hand, to avoid over-allocation or under-allocation of resources, we focus on solving the problem of fairness and energy efficiency balance of the Metaverse communication system. By continuously optimizing the EE of the minimum EE cell, we can realize the high EE of the total communication system while taking into account the EE fairness of each cell. On the other hand, previous works typically consider the classical RIS, which involves managing RIS components in a homogeneous grid arrangement with uniform cell distance. Due to the capability limitations of traditional RIS heavily depend on the quantity of RIS components, CSI acquisition cost will be enormous for massive RIS components. We concentrate on promoting the capacity and EE of the RIS-assisted systems model with finite RIS components. Motivated by the superiority of sparse phased array [34], [35], an irregular RIS frame is employed to elevate the capacity and EE of the system, which is distinguish from the existing regular RIS. The complexity of RIS controller is reduced by adopting irregular RIS to connect fewer RIS components, which is more consistent with the requirements of Metaverse service.

B. Contributions

The contributions are as follows:

- Firstly, this paper proposes a multi-UAV heterogeneous networks in Metaverse based on imperfect CSI, which includes an MBS, multiple UAVs and multiple irregular RISs to establish virtual communication links. Based on outage probability constraint, UAV transmit power constraint, Metaverse users' QoS, cross-layer interference and RIS phase constraint, a collaborative low complexity design of power control, RIS topology and reflecting phase design is proposed to realize system EE of the minimum UAV cell maximization.
- Secondly, the proposed joint design problem can be solved in three sub-stages. To maximize the EE of RIS auxiliary Metaverse service under the finite number of RIS components, a tabu search approach is employed to explore the optimal topology of irregular RIS. This

is the first attempt to apply the algorithm to RIS-UAV network optimization problem based on imperfect CSI. Then, the rate outage probability restriction is approximated through Bernstein type inequality. Under the rigid restrictions of the maximum transmit power of UAVs, Metaverse users' QoS requirements and cross-layer interference, a valid solution is received by invoking Bernstein-based algorithm. By solving the problem of RIS topology design and power control, the fairness EE optimization problem is transformed into a simplified form that only considers phase shift optimization. To tackle this subproblem, a low-complexity strategy that employs penalty is implemented.

- Lastly, the simulation curves exhibit that the joint optimization scheme has the advantages of fewer iterations and faster convergence. Through massive comparative simulation experiments, the scheme demonstrates a substantial enhancement in system EE, which exhibits remarkable effectiveness of the scheme. In addition, the deployment of irregular RIS has a tremendous impact on the improvement of capacity and system EE. What's more, the low complexity RIS controller design combining phase shift design and topology design of irregular RIS plays a prominent role in enhancing the EE for Metaverse.

C. Organization

The remaining sections are arranged as follows. Section II establishes the fairness model of system EE. In Section III, the three parameters of irregular RIS topology, power control and reflecting phase is optimized in turn. In Section IV, simulations and comparative experiments are exhibited to certify the superiority of the proposed scheme. Lastly, Section V elaborates the conclusion.

D. Notations

$\mathbf{\Lambda}^H$ denotes the conjugate transpose of vector $\mathbf{\Lambda}$. $\|\mathbf{\Lambda}\|$, $\|\mathbf{\Lambda}\|_F$, $\|\mathbf{\Lambda}\|_*$ are the Euclidean norm, Frobenius norm, nuclear norm of matrix $\mathbf{\Lambda}$, respectively. $\mathbf{\Lambda} \succeq 0$ represents $\mathbf{\Lambda}$ as a positive semi-definite matrix. $\text{rank}(\cdot)$ and $\text{tr}(\cdot)$ are the rank and trace of matrix, respectively. $(\cdot)^*$ is the optimal solution for the parameter. $\text{Re}\{\cdot\}$ denotes the real part of a complex number.

II. SYSTEM MODEL AND PROBLEM FORMULATION

A. System Model

In Fig. 1, this paper proposes a cross-layer multi-cell downlink system in Metaverse, which consists of one ground macro cell and K UAV cells. The k -th UAV cell consists of U_k Metaverse users, denoted as UAV user equipment (UUE) and a UAV base station, denoted as UBS(k). Each UAV cell is equipped with RIS devices to assist Metaverse users to enhance signal reception. L RIS devices are distributed in the system, and the elements of the RIS are arranged irregularly. The $(K+1)$ -th BS is labeled as macro base station (MBS) for serving Metaverse macro users (MUE). Both base stations

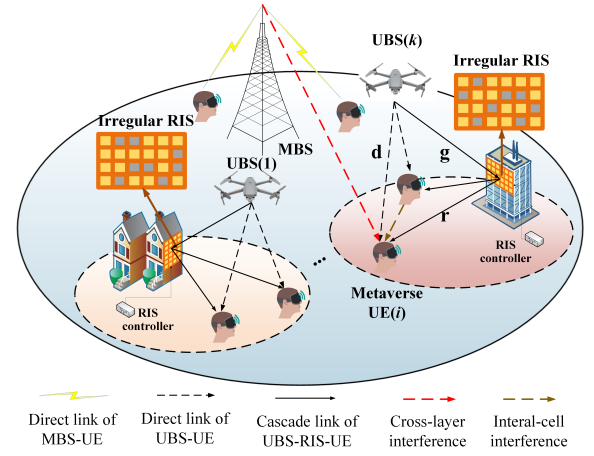


Fig. 1. Irregular RIS-enhanced multi-UAV communication system in Metaverse.

and metaverse users in the system are equipped with a single antenna. The i -th Metaverse user in the k -th UAV unit is marked as UUE(k, i), and the corresponding position coordinates are $\mathbf{u}_{k,i} = [\mathbf{u}_{k,i}^x, \mathbf{u}_{k,i}^y]^T$. The position coordinates of the k -th UAV are $\mathbf{q}_k = [\mathbf{q}_k^x, \mathbf{q}_k^y, \mathbf{q}_k^z]^T$. The ℓ -th RIS is recorded as RIS(ℓ), and the position coordinates are $\mathbf{I}_\ell = [\mathbf{I}_\ell^x, \mathbf{I}_\ell^y, \mathbf{I}_\ell^z]^T$, $\ell \in L$.

This paper presents an irregular RIS, whose array is regarded as a uniformly divided grid point. By selecting the feasible positions of the RIS elements from all the grid points, the irregular RIS may implement equivalently. The RIS in this paper is equipped with uniform linear array (ULA) and the results can be directly extended to other antenna topologies, such as uniform planar array (UPA). For convenience, this paper considers a fixed network spacing, in which M reflecting components are sparsely distributed on the grid points of the magnified surface M_s , $M_s > M$, half of carrier frequency wavelength is defined as the length between neighboring mesh points. $\mathbf{\Theta}_\ell = \text{diag}(\theta_1^\ell, \dots, \theta_m^\ell, \dots, \theta_{M_s}^\ell)$ represents the phase matrix of RIS(ℓ), θ_m^ℓ is the reflection coefficient of the m -th component on RIS(ℓ), $\theta_m^\ell = e^{j\varphi_m^\ell}$, φ_m^ℓ indicates the phase of the m -th component on RIS(ℓ). $\theta_\ell = [\theta_1^\ell, \dots, \theta_m^\ell, \dots, \theta_{M_s}^\ell]^H$, $[\cdot]^H$ is the conjugate transpose of vector $[\cdot]$. Regardless of multiple reflections, the topology selection matrix of RIS(ℓ) is $\mathbf{X}_\ell = \text{diag}(\mathbf{x}_\ell)$, $\mathbf{x}_\ell = [x_1^\ell, x_2^\ell, \dots, x_{M_s}^\ell]^T$. Let $x_m^\ell \in \{0, 1\}$, "1" represents selection, "0" represents opposite, corresponding to whether to deploy the RIS component at the m -th grid point, respectively.

Considering the LoS and non-line-of-sight (NLoS) propagate signals in the downlink of multi-UAV cells in Metaverse, the channel gains of the three links are expressed in the following [36].

The direct channel gain between UBS(k) and UUE(k, i) can be expressed

$$\hat{d}_{k,i} = \sqrt{\frac{\rho_0}{\|\mathbf{q}_k - \mathbf{u}_{k,i}\|^{\alpha_{dl}}}} \left(\sqrt{\frac{\kappa_{dl}}{\kappa_{dl} + 1}} \bar{d}_{k,i} + \sqrt{\frac{1}{\kappa_{dl} + 1}} \bar{\bar{d}}_{k,i} \right) \quad (1)$$

where ρ_0 represents the path-loss at 1 m reference range, κ_{dl} indicates Rician factor, α_{dl} represents the path-loss index.

$\bar{d}_{k,i,c} = 1$ represents the deterministic LoS link component, $\bar{d}_{k,i,c}$ is random Rayleigh distribution NLoS components.

Similarly, the cascaded channel gains of UBS(k)-RIS(ℓ) and RIS(ℓ)-UUE(k, i) can be expressed

$$\hat{\mathbf{g}}_k^\ell = \sqrt{\frac{\rho_0}{\|\mathbf{q}_k - \mathbf{I}_\ell\|^2}} \left[1, e^{-j\frac{2\pi\varrho}{\lambda} \cos \varphi_k^\ell}, \dots, e^{-j\frac{2\pi(M_s-1)\varrho}{\lambda} \cos \varphi_k^\ell} \right]^T \quad (2)$$

$$\hat{\mathbf{r}}_{k,i}^\ell = \sqrt{\frac{\rho_0}{\|\mathbf{I}_\ell - \mathbf{u}_{k,i}\|^{\alpha_{rl}}}} \left(\sqrt{\frac{\kappa_{rl}}{\kappa_{rl} + 1}} \bar{\mathbf{r}}_{k,i}^\ell + \sqrt{\frac{1}{\kappa_{rl} + 1}} \bar{\mathbf{r}}_{k,i}^\ell \right) \quad (3)$$

where $\cos \varphi_k^\ell = \frac{\mathbf{I}_\ell - \mathbf{q}_k^x}{\|\mathbf{I}_\ell - \mathbf{q}_k\|}$ represents the cosine of the angle-of-arrival (AoA) from UBS(k) to RIS(ℓ), $\mathbf{I}_\ell^x$ is the coordinate value in the x -axis of the ℓ -th RIS position coordinate, $\mathbf{q}_k^x$ is the coordinate value in the x -axis of the k -th UAV position coordinate. ϱ indicates the unit space of RIS, λ denotes the carrier wavelength, κ_{rl} is Rician factor, α_{rl} denotes the path-loss index. $\bar{\mathbf{r}}_{k,i}^\ell$ is random Rayleigh distribution NLoS components. $\bar{\mathbf{r}}_{k,i}^\ell$ is the deterministic LoS component of RIS(ℓ)-UUE(k, i), expressed as

$$\bar{\mathbf{r}}_{k,i}^\ell = \left[1, e^{-j\frac{2\pi\varrho}{\lambda} \cos \psi_{k,i}^\ell}, \dots, e^{-j\frac{2\pi(M_s-1)\varrho}{\lambda} \cos \psi_{k,i}^\ell} \right]^T \quad (4)$$

where $\cos \psi_{k,i}^\ell = \frac{\mathbf{u}_{k,i}^x - \mathbf{I}_\ell^x}{\|\mathbf{u}_{k,i} - \mathbf{I}_\ell\|}$ indicates the cosine of the angle-of-departure (AoD) from RIS(ℓ) to UUE(k, i), $\mathbf{u}_{k,i}^x$ is the coordinate value in the x -axis of the position coordinate of UUE(k, i).

Employing RIS in Metaverse communication, the united channel gain of UBS(k) to UUE(k, i) link is

$$\hat{h}_{k,i}^\ell = \hat{d}_{k,i}^H + (\hat{\mathbf{r}}_{k,i}^\ell)^H \mathbf{X}_\ell \Theta_\ell \hat{\mathbf{g}}_k^\ell = \hat{d}_{k,i}^H + \theta_\ell^H \hat{\mathbf{G}}_{k,i}^\ell \quad (5)$$

where $\hat{\mathbf{G}}_{k,i}^\ell = \text{diag}[(\hat{\mathbf{r}}_{k,i}^\ell)^H \mathbf{X}_\ell] \hat{\mathbf{g}}_k^\ell$ represents the joint equivalent cascaded channel gain from UBS(k) through RIS(ℓ) to UUE(k, i). The cascaded channel gain has the following deformation, which gives priority to the statistical CSI error model

$$\mathbf{G}_{k,i}^\ell = \hat{\mathbf{G}}_{k,i}^\ell + \Delta \mathbf{G}_{k,i}^\ell \quad (6)$$

where $\Delta \mathbf{G}_{k,i}^\ell$ represents the unknown estimation error of the cascaded chain, which obeys the distribution of circularly symmetric complex Gaussian (CSCG). $\text{vec}(\Delta \mathbf{G}_{k,i}^\ell) \sim \mathcal{CN}(\mathbf{0}, \Sigma_{k,i}^\ell)$, where $\Sigma_{k,i}^\ell \succeq \mathbf{0}$.

Since non-orthogonal multiple access (NOMA) permits multi-users to occupy one subchannel and the receiving signals may be decoded by successive interference elimination, NOMA can reduce interference in certain orders depending on the channel gain or power of different Metaverse users. The user's decoding order depends on the order in which the channel gains grow. Assuming that within a certain period of time, the decoding sequence of Metaverse users in the k -th cell is $|h_{k,1}^\ell|^2 \leq |h_{k,2}^\ell|^2 \leq \dots \leq |h_{k,U_k}^\ell|^2$, that is to say, the optimal Metaverse user decoding sequence is identified by the effective channel gain and arranged in descending order.

Considering the interference of other UUEs in the cell and the interference of MUEs, the interference-plus-noises power

of UUE(k, i) is

$$\beta_{k,i}^\ell = |h_{k,k,i}^\ell|^2 \sum_{j=i+1}^{U_k} p_{k,j}^\ell + |h_{k,K+1,i}^\ell|^2 \sum_{v=1}^{U_{K+1}} p_v + \sigma^2 \quad (7)$$

where σ^2 represents the power of additive white Gaussian noise (AWGN), $i, j \in U_k$, p_v is the power beam of MUE(v). $p_{k,i}^\ell$ is the power of UUE(k, i) containing RIS ℓ . Let $h_{k,k,i}^\ell$ is the equivalent channel form from UBS(k) to UUE(k, i) in the cell containing RIS ℓ , $h_{k,K+1,i}^\ell$ signifies the equivalent channel from MBS to UUE(k, i) in the cell containing RIS ℓ .

Hence the received signal-to-noise ratio of UUE(k, i) is

$$\text{SINR}_{k,i}^\ell = \frac{p_{k,i}^\ell}{\beta_{k,i}^\ell} |h_{k,k,i}^\ell|^2 \quad (8)$$

B. Problem Formulation

This paper focuses on RIS topology design, UAV transmit power optimization, and RIS reflecting phase design to achieve fairness by maximizing system EE of the minimum UAV cell.

The scheduled data rate of UUE(k, i) is

$$\begin{aligned} \hat{R}_{k,i}^\ell &= B_{sc} \log_2 \left(1 + \frac{|h_{k,k,i}^\ell|^2 p_{k,i}^\ell}{|h_{k,k,i}^\ell|^2 \sum_{j=i+1}^{U_k} p_{k,j}^\ell + |h_{k,K+1,i}^\ell|^2 \sum_{v=1}^{U_{K+1}} p_v + \sigma^2} \right) \end{aligned} \quad (9)$$

where B_{sc} is the channel bandwidth. Considering the imperfect CSI in the communication link, the achievable capacity of UUE(k, i) is

$$R_{k,i}^\ell = B_{sc} \log_2 (1 + \text{SINR}_{k,i}^\ell) \quad (10)$$

The sum-rate of all UAV cells are

$$R(\mathbf{X}, \mathbf{P}, \Theta) = \sum_{k=1}^K \sum_{i=1}^{U_k} R_{k,i}^\ell \quad (11)$$

where $\mathbf{X} = \{x_\ell, \forall \ell\}$ is topology matrix of RISs. Since each UAV cell has different user locations, the RIS topology is different for each cell. $\mathbf{P} = \{p_{k,i}^\ell, \forall k, i, \ell\}$ is transmit power matrix. $\Theta = \{\theta_m^\ell, \forall m, \ell\}$ denotes the phase shift matrix of RISs.

The whole UAV cells power consumption is

$$T(\mathbf{X}, \mathbf{P}, \Theta) = \sum_{k=1}^K \sum_{i=1}^{U_k} p_{k,i}^\ell + P_{RIS} + P_C \quad (12)$$

where $P_{RIS} = LMp_{ris}$, P_{RIS} represents the power consumption of the RIS components, p_{ris} is the energy consumption of each RIS unit element [37]. $P_C = Kp_c$, p_c is the mechanical energy loss of the UAV to overcome gravitational hover [38].

The whole EE of UAV cells is formulated as

$$\text{EE}(\mathbf{X}, \mathbf{P}, \Theta) = \frac{R(\mathbf{X}, \mathbf{P}, \Theta)}{T(\mathbf{X}, \mathbf{P}, \Theta)} = \frac{\sum_{k=1}^K \sum_{i=1}^{U_k} R_{k,i}^\ell}{\sum_{k=1}^K \sum_{i=1}^{U_k} p_{k,i}^\ell + P_{RIS} + P_C} \quad (13)$$

The EE of the k -th UAV cell is

$$EE_k(\mathbf{X}, \mathbf{P}, \Theta) = \frac{R_k(\mathbf{X}, \mathbf{P}, \Theta)}{T_k(\mathbf{X}, \mathbf{P}, \Theta)} = \frac{\sum_{i=1}^{U_k} R_{k,i}^\ell}{\sum_{i=1}^{U_k} p_{k,i}^\ell + Mp_{ris} + p_c} \quad (14)$$

This paper aims at maximizing the minimum UAV cells' EE to attain fairness, thereby promoting the whole system EE. According to the above established system model, the following joint problem is established

$$\begin{aligned} P : \max_{\mathbf{X}, \mathbf{P}, \Theta} \min_k EE_k(\mathbf{X}, \mathbf{P}, \Theta) \\ s.t. \quad & C1 : x_m^\ell \in \{0, 1\}, \forall m, \ell \\ & C2 : \mathbf{1}^T \mathbf{x}_\ell = M, \forall \ell \\ & C3 : p_{k,i}^\ell \geq 0, \forall k, i \\ & C4 : \sum_{i=1}^{U_k} p_{k,i}^\ell \leq P_k^{UAV}, \forall k, i \\ & C5 : \sum_{i=1}^{U_k} \hat{R}_{k,i}^\ell \geq R_{k,\min}, \forall k \\ & C6 : |h_{k,K+1}^\ell|^2 \sum_{i=1}^{U_k} p_{k,i}^\ell \leq \Delta_{\max} \\ & C7 : \phi[R_{k,i}^\ell \geq \hat{R}_{k,i}^\ell] \geq 1 - \alpha_{out}, \forall k, i \\ & C8 : |\theta_m^\ell| = 1, \forall m, \ell \end{aligned} \quad (15)$$

This paper optimizes the three parameters in turn to obtain the system EE. Among them, $C1$ and $C2$ are the topology design limits of the irregular RIS, where the value of a diagonal element of the RIS topology matrix $\mathbf{X}$ is 1, and the remaining diagonal elements $M_s - M$ is specified as 0. $C3$ and $C4$ are the constraints of the power control parameters. $C5$ is the Metaverse users' sum rate constraints. $C6$ is the interference limit of the macro cell to UUE [40], $C7$ guarantees that the probability of Metaverse users succeed in decoding its message at the rate $\hat{R}_{k,i}^\ell$ is not fewer than that $1 - \alpha_{out}$. $C8$ is the RIS phase shift constraints.

III. PROPOSED SCHEME

A. RIS Topology Design

The problem of RIS topology design can be transformed into

$$\begin{aligned} P(A) : \max_{\mathbf{x}_k} \min_k EE_k \\ s.t. \quad & C1 : x_m^\ell \in \{0, 1\}, \forall m, \ell \\ & C2 : \mathbf{1}^T \mathbf{x}_\ell = M, \forall \ell \end{aligned} \quad (16)$$

This paper assumes that the design topology of the RIS is fixed for a brief interval. That is, the topology design of the RIS requires only adaptively varies in long time scales. The cost of acquiring CSI is identical to that of a conventional RIS with an equal number of RIS components, as the corresponding CSI dimension of a specific RIS topology is connected to the chosen quantity of RIS components. This paper uses

Algorithm 1 Tabu-Based Algorithm for Solving P(A)

- 1: Initialize: Tabu table $Tabu_L = \emptyset$, Maximum value of tabuo list $Tabu_L^{MAX}$, Distance to nearby neighbors $Neigh_D$, Spatial extent of neighbors $Neigh_S$, Iteration number I_A , RIS topology matrix $\mathbf{X}_\ell^0$
- 2: Calculate the UAV cells with the minimum EE in the system, the location is marked as $\min EE_{Location}$.
- 3: **for** $i_A = 1$ to I_A **do**
- 4: Dynamically adjust $Neigh_D$ according to i_A . Randomly replace $Neigh_D$ 1 with 0 within the diagonal entries of $\mathbf{X}_\ell^{i_A}$, and create replacement neighbors of $\mathbf{X}_\ell^{i_A}$.
- 5: Eliminate the tabu neighbours in $Tabu_L$, get $Neigh_S$ candidates.
- 6: Calculate the corresponding EE for all $Neigh_S$ candidate neighbors. Then, we select EEmax as the best candidate and save it as $\mathbf{X}_\ell^{i_A+1}$.
- 7: **if** $Tabu_L < Tabu_L^{MAX}$ **then**
- 8: Add $\mathbf{X}_\ell^{i_A}$ to the $Tabu_L$
- 9: **else**
- 10: Add $\mathbf{X}_\ell^{i_A}$ to the end of $Tabu_L$ and eliminate the first element of $Tabu_L$
- 11: **end if**
- 12: **end for**

the Tabu search mechanism [39] to design the topology of irregular RIS.

First, initialize the RIS topology matrix. Second, find the cell with the minimum energy efficiency in all UAV cells and mark it. The initial topology is fixed, and the power control and phase shift parameters are updated by Algorithms 2 and 3, respectively. Finally, implement the tabu search technique for optimizing the topology variable of RIS. The key thought of the design is to acquire the locally optimized solution and mark it, avert the local optimization solution in the further iteration, and expand the search range. By comparing multiple local optimal solutions, the global optimal solution is obtained. An important concept in the tabu-search method is the neighborhood and the tabu list. In the algorithm design of this paper, a larger neighbor distance is selected at the beginning of the iteration to obtain a wider search range, and as the iteration progresses, the algorithm will reduce the neighbor distance for dynamic adjustment. As the iteration progresses, the tabu table is continuously updated. Through multiple iterations, the earliest move that enters the tabu table will be released from the tabu table. The acquired neighbors are inspected in the tabu list, avoiding multiple searches of a topology. The specific algorithm details are as in Algorithm 1.

B. UBS Power Control

Since the topology matrix of the RIS has been obtained in the previous subsection, the topology matrix is regarded as a constant, the phase matrix of RISs are fixed, so the power allocation subproblem in heterogeneous networks is

stated as

$$\begin{aligned}
P(B) : \max_{\mathbf{p}_k} \min_k EE_k \\
s.t. \quad C3 : \sum_{i=1}^{U_k} p_{k,i}^\ell \leq P_k^{UAV}, \forall k, i \\
C4 : \sum_{i=1}^{U_k} R_{k,i}^\ell \geq R_{k,\min}, \forall k \\
C5 : p_{k,i}^\ell \geq 0, \forall k, i \\
C6 : |h_{k,K+1}^\ell|^2 \sum_{i=1}^{U_k} p_{k,i}^\ell \leq \Delta_{\max} \\
C7 : \phi[R_{k,i}^\ell \geq \hat{R}_{k,i}^\ell] \geq 1 - \alpha_{out}, \forall k, i \quad (17)
\end{aligned}$$

Apparently, the target form and restrictions are both non-convex functions and non-convex constraints. We first apply for the fractional programming theory to convert the nonlinear fractional form of the target form into subtraction. Introducing the variable Π , the target has the following form

$$\Pi^* = \max_{\mathbf{p}_k} \min_k \frac{R_k(\mathbf{p}_k)}{T_k(\mathbf{p}_k)} = \frac{R(\mathbf{P}^*)}{T(\mathbf{P}^*)} \quad (18)$$

The auxiliary variable ϖ is introduced to further simplify the objective function as follows

$$\begin{aligned}
\max_{\mathbf{p}_k} \varpi \\
s.t. \quad C9 : R_k(\mathbf{p}_k) - \Pi T_k(\mathbf{p}_k) \geq \varpi \quad (19)
\end{aligned}$$

Let $P_{k,i}^\ell = (p_{k,i}^\ell)^{1/2}((p_{k,i}^\ell)^H)^{1/2}$. For $R_{k,i}^\ell$, the bisection is formalized as a subtraction

$$\begin{aligned}
R_{k,i}^\ell \\
= B_{sc} \log_2 \left(1 + \frac{|h_{k,k,i}^\ell|^2 p_{k,i}^\ell}{|h_{k,k,i}^\ell|^2 \sum_{j=i+1}^{U_k} p_{k,j}^\ell + |h_{k,K+1,i}^\ell|^2 \sum_{\nu=1}^{U_{K+1}} p_{\nu} + \sigma^2} \right) \\
= B_{sc} (f_1(P_{k,i}^\ell) - f_2(P_{k,i}^\ell)) \quad (20)
\end{aligned}$$

where $\text{tr}(\cdot)$ represents the trace of matrix $(\cdot)$.

$$\begin{aligned}
f_1(P_{k,i}^\ell) \\
= \log_2 \left(\sum_{j=i}^{U_k} \text{tr}(H_{k,k,i}^\ell P_{k,j}^\ell) + \sum_{\nu=1}^{U_{K+1}} \text{tr}(H_{k,K+1,i}^\ell P_\nu) + \sigma^2 \right) \quad (21)
\end{aligned}$$

$$\begin{aligned}
f_2(P_{k,i}^\ell) \\
= \log_2 \left(\sum_{j=i+1}^{U_k} \text{tr}(H_{k,k,i}^\ell P_{k,j}^\ell) + \sum_{\nu=1}^{U_{K+1}} \text{tr}(H_{k,K+1,i}^\ell P_\nu) + \sigma^2 \right) \quad (22)
\end{aligned}$$

Since $f_2(P_{k,i}^\ell)$ is a differentiable convex function, the following deformation is obtained on the basis of the first-order Taylor expansion

$$\begin{aligned}
f_2(P_{k,i}^\ell) \leq \text{tr}((\nabla_P^H f_2(P_{k,i}^\ell)^{(t-1)})(P_{k,i}^\ell - (P_{k,i}^\ell)^{(t-1)})) \\
+ f_2(P_{k,i}^\ell)^{(t-1)} \triangleq \tilde{f}_2(P_{k,i}^\ell) \quad (23)
\end{aligned}$$

where

$$\begin{aligned}
\nabla_P^H f_2(P_{k,i}^\ell)^{(t-1)} \\
= \frac{H_{k,k,i}^\ell / \ln 2}{\sum_{j=i+1}^{U_k} \text{tr}(H_{k,k,i}^\ell (P_{k,j}^\ell)^{t-1}) + \sum_{\nu=1}^{U_{K+1}} \text{tr}(H_{k,K+1,i}^\ell (P_\nu)^{t-1}) + \sigma^2} \quad (24)
\end{aligned}$$

The Metaverse user's outage probability can be rewritten as Eq(25), shown at the bottom of the next

page, where $\chi_{k,i}^\ell = \sum_{\nu=1}^{U_{K+1}} \text{tr}(H_{k,K+1,i}^\ell P_\nu) + \sigma^2$, $Z_{k,i}^\ell = P_{k,i}^\ell / (2^{\hat{R}_{k,i}^\ell / B_{sc}} - 1) - \sum_{j=i+1}^{U_k} P_{k,j}^\ell$.

Let $\Phi_\ell = \theta_\ell \theta_\ell^H$, $\Sigma_{k,i}^\ell = (\lambda_{k,i}^\ell)^2 \mathbf{E}$, then the CSI error vector is denoted as $\text{vec}(\Delta \mathbf{G}_{k,i}^\ell) = \lambda_{k,i}^\ell \mathbf{e}_{k,i}^\ell$, $\mathbf{e}_{k,i}^\ell \sim \mathcal{CN}(\mathbf{0}, \mathbf{E})$, then Eq. (25) can be convert as Eq. (26), shown at the bottom of the next page.

So C7 can be remolded as

$$\begin{aligned}
\phi \left[(\mathbf{e}_{k,i}^\ell)^H \mathbf{Y}_{k,i}^\ell \mathbf{e}_{k,i}^\ell + 2\text{Re} \left\{ (\mathbf{y}_{k,i}^\ell)^H \mathbf{e}_{k,i}^\ell \right\} + y_{k,i}^\ell \geq 0 \right] \\
\geq 1 - \alpha_{out} \quad (27)
\end{aligned}$$

where

$$\begin{aligned}
\mathbf{Y}_{k,i}^\ell &= (\lambda_{k,i}^\ell)^2 (Z_{k,i}^\ell \Phi_\ell) \\
\mathbf{y}_{k,i}^\ell &= \lambda_{k,i}^\ell \text{vec}((\theta_\ell d_{k,i}^H + \Phi_\ell \hat{\mathbf{G}}_{k,i}^\ell) Z_{k,i}^\ell) \\
y_{k,i}^\ell &= (d_{k,i}^H + \theta_\ell^H \hat{\mathbf{G}}_{k,i}^\ell) Z_{k,i}^\ell (d_{k,i} + (\hat{\mathbf{G}}_{k,i}^\ell)^H \theta_\ell) - \chi_{k,i}^\ell \quad (28)
\end{aligned}$$

With the help of auxiliary variable $\mu_{k,i}^\ell$, according to the Bernstein-Type inequality [41], Eq.(27) can be further simplified as

$$\begin{aligned}
C7'(a) : \text{tr}(\mathbf{Y}_{k,i}^\ell) - \sqrt{2 \ln(1/\alpha_{out})} \mu_{k,i}^\ell \\
+ \ln(\alpha_{out}) \varepsilon_{k,i}^\ell + y_{k,i}^\ell \geq 0 \\
C7'(b) : \sqrt{\|\mathbf{Y}_{k,i}^\ell\|_F^2 + 2\|\mathbf{y}_{k,i}^\ell\|^2} \leq \mu_{k,i}^\ell \\
C7'(c) : \varepsilon_{k,i}^\ell \geq 0 \\
C7'(d) : \varepsilon_{k,i}^\ell \mathbf{E} + \mathbf{Y}_{k,i}^\ell \succeq 0 \quad (29)
\end{aligned}$$

where

$$\text{tr}(\mathbf{Y}_{k,i}^\ell) = (\lambda_{k,i}^\ell)^2 \text{tr}(Z_{k,i}^\ell \Phi_\ell) = (\lambda_{k,i}^\ell)^2 Z_{k,i}^\ell M \quad (30)$$

$$\|\mathbf{Y}_{k,i}^\ell\|_F^2 = (\lambda_{k,i}^\ell)^4 \|\mathbf{Z}_{k,i}^\ell \Phi_\ell\|_F^2 = (\lambda_{k,i}^\ell)^4 (M Z_{k,i}^\ell)^2 \quad (31)$$

$$\begin{aligned}
\|\mathbf{y}_{k,i}^\ell\|^2 &= (\lambda_{k,i}^\ell)^2 \|\text{vec}((\theta_\ell d_{k,i}^H + \Phi_\ell \hat{\mathbf{G}}_{k,i}^\ell) Z_{k,i}^\ell)\|^2 \\
&= (\lambda_{k,i}^\ell)^2 M \|(d_{k,i}^H + \theta_\ell^H \hat{\mathbf{G}}_{k,i}^\ell) Z_{k,i}^\ell\|_2^2 \quad (32)
\end{aligned}$$

$$\xi(\mathbf{Y}_{k,i}^\ell) = \xi((\lambda_{k,i}^\ell)^2 (Z_{k,i}^\ell \Phi_\ell)) = (\lambda_{k,i}^\ell)^2 Z_{k,i}^\ell M \quad (33)$$

where $\xi(\cdot)$ is the eigenvalue of the matrix. Using the above transformation process, $C7'(a)$ can be expressed as

$$(\lambda_{k,i}^\ell)^2 Z_{k,i}^\ell M - \sqrt{2 \ln(1/\alpha_{out})} \mu_{k,i}^\ell + \ln(\alpha_{out}) \varepsilon_{k,i}^\ell + y_{k,i}^\ell \geq 0 \quad (34)$$

Algorithm 2 Bernstein-Based Algorithm for Solving P(B)

-
- 1: Initialize the maximum number of iterations I_B , power matrix $\mathbf{p}_k^0$.
 - 2: **for** $i_B = 1$ to I_B **do**
 - 3: Solve Eq. (37) to obtain $\mathbf{P}_k^{i_B}$.
 - 4: Update $i_B = i_B + 1$.
 - 5: Update ϖ .
 - 6: **end for**
-

Convert $C7'(b)$ further as follows

$$\begin{aligned} & \sqrt{(\lambda_{k,i}^\ell)^4 (MZ_{k,i}^\ell)^2 + 2(\lambda_{k,i}^\ell)^2 M \|(d_{k,i}^H + \theta_\ell^H \hat{\mathbf{G}}_{k,i}^\ell) Z_{k,i}^\ell\|_2^2} \\ &= \left\| \begin{pmatrix} \lambda_{k,i}^\ell \\ \sqrt{2M} \lambda_{k,i}^\ell Z_{k,i}^\ell \left(d_{k,i} + (\hat{\mathbf{G}}_{k,i}^\ell)^H \theta_\ell \right) \end{pmatrix} \right\| \leq \mu_{k,i}^\ell \quad (35) \end{aligned}$$

Since the positive semidefinite of the matrix can be transformed into a non-negative eigenvalue problem, $\varepsilon_{k,i}^\ell \mathbf{E} + \mathbf{Y}_{k,i}^\ell$ can be converted to

$$\begin{aligned} \zeta(\varepsilon_{k,i}^\ell \mathbf{E} + \mathbf{Y}_{k,i}^\ell) &= \varepsilon_{k,i}^\ell \zeta(\mathbf{E}) + \zeta(\mathbf{Y}_{k,i}^\ell) \\ &= \varepsilon_{k,i}^\ell + (\lambda_{k,i}^\ell)^2 M Z_{k,i}^\ell \geq 0 \quad (36) \end{aligned}$$

Finally, problem P(B) can be reconstructed as follows

$$\begin{aligned} P'(B) : \max_{\mathbf{p}_k} \quad & \varpi \\ \text{s.t.} \quad & C3, C5 \\ & C4' : \sum_{i=1}^{U_k} B_{sc}(f_1(P_{k,i}^\ell) - \tilde{f}_2(P_{k,i}^\ell)) \geq R_{k,\min}, \forall k \\ & C6' : \sum_{i=1}^{U_k} \text{tr}(P_{k,i}^\ell H_{k,K+1}^\ell) \leq \Delta_{\max} \\ & C7'(a') : (\lambda_{k,i}^\ell)^2 Z_{k,i}^\ell M - \sqrt{2 \ln(1/\alpha_{out})} \mu_{k,i}^\ell \\ & \quad + \ln(\alpha_{out}) \varepsilon_{k,i}^\ell + y_{k,i}^\ell \geq 0 \end{aligned}$$

$$\begin{aligned} C7'(b') : & \left\| \begin{pmatrix} (\lambda_{k,i}^\ell)^2 M Z_{k,i}^\ell \\ \sqrt{2M} \lambda_{k,i}^\ell Z_{k,i}^\ell (d_{k,i} + (\hat{\mathbf{G}}_{k,i}^\ell)^H \theta_\ell) \end{pmatrix} \right\| \\ & \leq \mu_{k,i}^\ell \\ C7'(c') : & \varepsilon_{k,i}^\ell \geq 0 \\ C7'(d') : & \varepsilon_{k,i}^\ell + (\lambda_{k,i}^\ell)^2 M Z_{k,i}^\ell \geq 0 \\ C9' : & \sum_{i=1}^{U_k} B_{sc}(f_1(P_{k,i}^\ell) - \tilde{f}_2(P_{k,i}^\ell)) \\ & \quad - \Pi \left(\sum_{i=1}^{U_k} \text{tr}(P_{k,i}^\ell) + M p_{ris} + p_c \right) \geq \varpi \quad (37) \end{aligned}$$

The aforementioned subproblem exhibits convex and may invoke CVX solver such as Mosek to solve it [42]. The power control optimization course is summarized as Algorithm 2.

C. RIS Phase Shift Optimization

With fixed power control and RIS topology matrix parameters, the problem of RIS phase shift design is reconstructed as

$$\begin{aligned} P(C) : \max_{\Theta} \min_k \quad & EE_k \\ \text{s.t.} \quad & C5 : \sum_{i=1}^{U_k} \hat{R}_{k,i}^\ell \geq R_{k,\min}, \forall k \\ & C7 : \phi[R_{k,i}^\ell \geq \hat{R}_{k,i}^\ell] \geq 1 - \alpha_{out}, \forall k, i \\ & C8 : |\theta_m^\ell| = 1, \forall m, \ell \quad (38) \end{aligned}$$

Let $\tilde{\Phi}_\ell = \tilde{\theta}_\ell \tilde{\theta}_\ell^H$, $\tilde{\theta}_\ell = [\theta_\ell \ 1]^T$, C8 can be converted into

$$\begin{aligned} C8'(a) : & \tilde{\Phi}_\ell \succeq 0 \\ C8'(b) : & \text{rank}(\tilde{\Phi}_\ell) = 1 \\ C8'(c) : & \text{diag}(\tilde{\Phi}_\ell) = \mathbf{1}_{M+1} \quad (39) \end{aligned}$$

$$\begin{aligned} \phi[R_{k,i}^\ell \geq \hat{R}_{k,i}^\ell] &= \phi[B_{sc} \log_2(1 + \frac{|h_{k,k,i}^\ell|^2 p_{k,i}^\ell}{|h_{k,k,i}^\ell|^2 \sum_{j=i+1}^{U_k} p_{k,j}^\ell + |h_{k,K+1,i}^\ell|^2 \sum_{\nu=1}^{U_{K+1}} p_\nu + \sigma^2})] \geq \hat{R}_{k,i}^\ell] \\ &= \phi \left[h_{k,k,i}^\ell Z_{k,i}^\ell (h_{k,k,i}^\ell)^H - \sum_{\nu=1}^{U_{K+1}} \text{tr}(H_{k,K+1,i}^\ell P_\nu) - \sigma^2 \geq 0 \right] = \phi \left[h_{k,k,i}^\ell Z_{k,i}^\ell (h_{k,k,i}^\ell)^H - \chi_{k,i}^\ell \geq 0 \right] \quad (25) \end{aligned}$$

$$\begin{aligned} & \phi \left[h_{k,k,i}^\ell Z_{k,i}^\ell (h_{k,k,i}^\ell)^H - \chi_{k,i}^\ell \geq 0 \right] \\ &= \phi \left[\text{vec}^H(\Delta \mathbf{G}_{k,i}^\ell) (Z_{k,i}^\ell \Phi_\ell) \text{vec}(\Delta \mathbf{G}_{k,i}^\ell) + 2\text{Re} \left\{ \text{vec}^H \left((d_{k,i}^H \theta_\ell + \Phi_\ell \hat{\mathbf{G}}_{k,i}^\ell) Z_{k,i}^\ell \right) \text{vec}(\Delta \mathbf{G}_{k,i}^\ell) \right\} \right. \\ & \quad \left. + (d_{k,i}^H + \theta_\ell^H \hat{\mathbf{G}}_{k,i}^\ell) Z_{k,i}^\ell \left(d_{k,i} + (\hat{\mathbf{G}}_{k,i}^\ell)^H \theta_\ell \right) - \chi_{k,i}^\ell \geq 0 \right] \\ &= \phi \left[(\lambda_{k,i}^\ell)^2 (\mathbf{e}_{k,i}^\ell)^H (Z_{k,i}^\ell \Phi_\ell) \mathbf{e}_{k,i}^\ell + 2\text{Re} \left\{ \lambda_{k,i}^\ell \text{vec}^H \left((\theta_\ell d_{k,i}^H + \Phi_\ell \hat{\mathbf{G}}_{k,i}^\ell) Z_{k,i}^\ell \right) \mathbf{e}_{k,i}^\ell \right\} \right. \\ & \quad \left. + (d_{k,i}^H + \theta_\ell^H \hat{\mathbf{G}}_{k,i}^\ell) Z_{k,i}^\ell \left(d_{k,i} + (\hat{\mathbf{G}}_{k,i}^\ell)^H \theta_\ell \right) - \chi_{k,i}^\ell \geq 0 \right] \quad (26) \end{aligned}$$

For other constraints, convert $R_{k,i}^\ell$ as follows

$$R_{k,i}^\ell = B_{sc} \log_2 \left(1 + \frac{\text{tr} \left(P_{k,i}^\ell \mathbf{\Omega}_{k,i}^\ell \right)}{\sum_{j=i+1}^{U_k} \text{tr} \left(P_{k,j}^\ell \mathbf{\Omega}_{k,j}^\ell \right) + \sum_{\nu=1}^{U_{K+1}} \text{tr} \left(H_{k,K+1,i}^\ell P_\nu \right) + \sigma^2} \right) = B_{sc} \left(f_3(\tilde{\Phi}_\ell) - f_4(\tilde{\Phi}_\ell) \right) \quad (40)$$

where

$$f_3(\tilde{\Phi}_\ell) = \log_2 \left(\sum_{j=i}^{U_k} \text{tr} \left(P_{k,j}^\ell \mathbf{\Omega}_{k,j}^\ell \right) + \sum_{\nu=1}^{U_{K+1}} \text{tr} \left(H_{k,K+1,i}^\ell P_\nu \right) + \sigma^2 \right) \quad (41)$$

$$f_4(\tilde{\Phi}_\ell) = \log_2 \left(\sum_{j=i+1}^{U_k} \text{tr} \left(P_{k,j}^\ell \mathbf{\Omega}_{k,j}^\ell \right) + \sum_{\nu=1}^{U_{K+1}} \text{tr} \left(H_{k,K+1,i}^\ell P_\nu \right) + \sigma^2 \right) \quad (42)$$

Based on the above formula, $\mathbf{\Omega}_{k,i}^\ell = (\mathbf{A}_{k,i}^\ell)^H \tilde{\Phi}_\ell \mathbf{A}_{k,i}^\ell$, $\mathbf{A}_{k,i}^\ell = [(\text{diag}(\mathbf{r}_{k,i}^H) \mathbf{g}_k^\ell)^T d_{k,i}^*]^T$. Utilizing the first-order Taylor expansion, the following form is obtained

$$f_4(\tilde{\Phi}_\ell) \leq \text{tr}((\nabla_{\tilde{\Phi}_\ell}^H f_4(\tilde{\Phi}_\ell)^{(t)})(\tilde{\Phi}_\ell - (\tilde{\Phi}_\ell)^{(t)})) + f_4((\tilde{\Phi}_\ell)^{(t)}) \triangleq \tilde{f}_4(\tilde{\Phi}_\ell) \quad (43)$$

where (44), as shown at the bottom of the next page.

Introduce slack variables $\gamma_k = [\gamma_{k,1}^\ell, \dots, \gamma_{k,i}^\ell, \dots, \gamma_{k,U_k}^\ell]^T$, the outage probability formula can be transformed into

$$\phi[R_{k,i}^\ell \geq \hat{R}_{k,i}^\ell] = \phi[h_{k,k,i}^\ell Z_{k,i}^\ell (h_{k,k,i}^\ell)^H - \chi_{k,i}^\ell - \gamma_{k,i}^\ell \geq 0] \quad (45)$$

Similarly, $y_{k,i}^\ell$ can have the following transformations

$$(y_{k,i}^\ell)^\Theta = (d_{k,i}^H + \theta_\ell^H \hat{\mathbf{G}}_{k,i}^\ell) Z_{k,i}^\ell (d_{k,i} + (\hat{\mathbf{G}}_{k,i}^\ell)^H \theta_\ell) - \chi_{k,i}^\ell - \gamma_{k,i}^\ell = \text{tr}(\mathbf{\Gamma}_{k,i}^\ell \tilde{\Phi}_\ell) + d_{k,i}^H Z_{k,i}^\ell d_{k,i} - \chi_{k,i}^\ell - \gamma_{k,i}^\ell \quad (46)$$

$$\text{where } \mathbf{\Gamma}_{k,i}^\ell = \begin{bmatrix} \hat{\mathbf{G}}_{k,i}^\ell Z_{k,i}^\ell (\hat{\mathbf{G}}_{k,i}^\ell)^H & \hat{\mathbf{G}}_{k,i}^\ell Z_{k,i}^\ell d_{k,i} \\ d_{k,i}^H Z_{k,i}^\ell (\hat{\mathbf{G}}_{k,i}^\ell)^H & 0 \end{bmatrix}.$$

Then $C7'(b')$ can be transformed as follows

$$\left\| \frac{(\lambda_{k,i}^\ell)^2 M Z_{k,i}^\ell}{\sqrt{2M} \lambda_{k,i}^\ell Z_{k,i}^\ell (d_{k,i} + (\hat{\mathbf{G}}_{k,i}^\ell)^H \theta_\ell)} \right\|^2 = (\lambda_{k,i}^\ell)^4 (M Z_{k,i}^\ell)^2 + 2M (\lambda_{k,i}^\ell)^2 \left\| (d_{k,i}^H + \theta_\ell^H \hat{\mathbf{G}}_{k,i}^\ell) Z_{k,i}^\ell \right\|_2^2 \quad (47)$$

where

$$\left\| (d_{k,i}^H + \theta_\ell^H \hat{\mathbf{G}}_{k,i}^\ell) Z_{k,i}^\ell \right\|_2^2$$

Algorithm 3 Penalty-Based Algorithm for Solving P(C)

- 1: Initialize maximum iterations I_C , phase matrix $\tilde{\Phi}_\ell^0$.
- 2: **for** $i_C = 1$ to I_C **do**
- 3: Solve Eq. (56) to obtain $\tilde{\Phi}_\ell^{t_C}$.
- 4: Update $i_C = i_C + 1$.
- 5: Update $EE_{\min_{\text{Location}}}$.
- 6: **end for**

$$= (d_{k,i}^H + \theta_\ell^H \hat{\mathbf{G}}_{k,i}^\ell) Z_{k,i}^\ell (Z_{k,i}^\ell)^* (d_{k,i} + (\hat{\mathbf{G}}_{k,i}^\ell)^H \theta_\ell) = \text{tr}(\mathbf{Q}_{k,i}^\ell \tilde{\Phi}_{k,i}^\ell) + d_{k,i}^H Z_{k,i}^\ell (Z_{k,i}^\ell)^* d_{k,i} \quad (48)$$

$$\mathbf{Q}_{k,i}^\ell = \begin{bmatrix} \hat{\mathbf{G}}_{k,i}^\ell Z_{k,i}^\ell (Z_{k,i}^\ell)^* (\hat{\mathbf{G}}_{k,i}^\ell)^H & \hat{\mathbf{G}}_{k,i}^\ell Z_{k,i}^\ell (Z_{k,i}^\ell)^* d_{k,i} \\ d_{k,i}^H Z_{k,i}^\ell (Z_{k,i}^\ell)^* (\hat{\mathbf{G}}_{k,i}^\ell)^H & 0 \end{bmatrix} \quad (49)$$

$C7'(b')$ can be simplified to

$$(\lambda_{k,i}^\ell)^4 M^2 (Z_{k,i}^\ell)^2 + 2M (\lambda_{k,i}^\ell)^2 \text{tr}(\mathbf{Q}_{k,i}^\ell \tilde{\Phi}_{k,i}^\ell) + d_{k,i}^* Z_{k,i}^\ell (Z_{k,i}^\ell)^* d_{k,i} \leq |\mu_{k,i}^\ell|^2 \quad (50)$$

For complex variables $\mu_{k,i}^\ell$, according to the first-order Taylor inequality, the following inequality is derived

$$(\mu_{k,i}^\ell)^2 \geq 2\text{Re} \left\{ (\mu_{k,i}^\ell)^{(n)} \mu_{k,i}^\ell \right\} - (\mu_{k,i}^\ell)^{(n),2} \quad (51)$$

where $(\cdot)^{(n)}$ represents the solution for the n -th iteration. Eq.(50) can be transformed into

$$(\lambda_{k,i}^\ell)^4 M^2 (Z_{k,i}^\ell)^2 + 2M (\lambda_{k,i}^\ell)^2 \text{tr}(\mathbf{Q}_{k,i}^\ell \tilde{\Phi}_{k,i}^\ell) + d_{k,i}^* Z_{k,i}^\ell (Z_{k,i}^\ell)^* d_{k,i} \leq 2\text{Re} \left\{ (\mu_{k,i}^\ell)^{(n)} \mu_{k,i}^\ell \right\} - (\mu_{k,i}^\ell)^{(n),2} \quad (52)$$

Hence, P(C) is reformulated as follows

$P'(C)$:

find $\tilde{\Phi}_\ell$

$$s.t. \quad C5': \sum_{i=1}^{U_k} B_{sc} (f_3(\tilde{\Phi}_\ell) - \tilde{f}_4(\tilde{\Phi}_\ell)) \geq R_{k,\min}, \forall k$$

$$C7''(a): (\lambda_{k,i}^\ell)^2 M Z_{k,i}^\ell - \sqrt{2 \ln(1/\alpha_{out})} \mu_{k,i}^\ell + \ln(\alpha_{out}) \varepsilon_{k,i}^\ell + (y_{k,i}^\ell)^\Theta \geq 0$$

$$C7''(b): (\lambda_{k,i}^\ell)^4 M^2 (Z_{k,i}^\ell)^2 + 2M (\lambda_{k,i}^\ell)^2 \text{tr}(\mathbf{Q}_{k,i}^\ell \tilde{\Phi}_{k,i}^\ell) + d_{k,i}^* Z_{k,i}^\ell (Z_{k,i}^\ell)^* d_{k,i} \leq 2\text{Re} \left\{ (\mu_{k,i}^\ell)^{(n)} \mu_{k,i}^\ell \right\} - (\mu_{k,i}^\ell)^{(n),2}$$

$$C7''(c): \gamma_{k,i}^\ell \geq 0$$

$$C8'(a): \tilde{\Phi}_\ell \succeq 0$$

$$C8'(b): \text{rank}(\tilde{\Phi}_\ell) = 1$$

$$C8'(c): \text{diag}(\tilde{\Phi}_\ell) = \mathbf{1}_{M+1}$$

$$C9'': \sum_{i=1}^{U_k} B_{sc} (f_3(\tilde{\Phi}_\ell) - \tilde{f}_4(\tilde{\Phi}_\ell)) - \Pi \left(\sum_{i=1}^{U_k} \text{tr}(P_{k,i}^\ell) + M p_{ris} + p_c \right) \geq \varpi \quad (53)$$

Obviously, the above problem is a feasibility problem. For $C8'(b)$ in the above problem, the penalty term is used to deal with the rank 1 problem [43]. The rank 1 problem can be equivalent to

$$\left\| \tilde{\Phi}_\ell \right\|_* - \left\| \tilde{\Phi}_\ell \right\|_2 \leq 0 \quad (54)$$

where $\left\| \tilde{\Phi}_\ell \right\|_* = \sum_{\tau} \sigma_\tau^\ell \geq \left\| \tilde{\Phi}_\ell \right\|_2 = \max_{\tau} \{\sigma_\tau^\ell\}$, σ_τ^ℓ is the τ -th singular value of $\tilde{\Phi}_\ell$, the equation is true precisely when $\tilde{\Phi}_\ell$ reaches rank 1, so the penalty-based problem is reconstructed as

$$P''(C) : \min_{s.t.} \Delta \left(\left\| \tilde{\Phi}_\ell \right\|_* - \left\| \tilde{\Phi}_\ell \right\|_2 \right) \quad C5', C7'', C9'', C8'(a), C8'(c) \quad (55)$$

where Δ is the penalty factor, the above problem is non-convex. For the above closed form of difference of concave functions, to deal with this problem, define a lower bound $B = \left\| \tilde{\Phi}_\ell \right\|_2$,

$$\begin{aligned} B(\tilde{\Phi}_\ell) &\geq B(\tilde{\Phi}_\ell^{(t)}) + \text{tr} \left(\nabla_{\tilde{\Phi}_\ell}^H B(\tilde{\Phi}_\ell^{(t)}) (\tilde{\Phi}_\ell - \tilde{\Phi}_\ell^{(t)}) \right) \\ &\triangleq \tilde{B}(\tilde{\Phi}_\ell) \end{aligned} \quad (56)$$

So the subproblem stated above is remolded as

$$P''(C) : \min_{s.t.} \Delta \left(\left\| \tilde{\Phi}_\ell \right\|_* - \tilde{B}(\tilde{\Phi}_\ell) \right) \quad C5', C7'', C9'', C8'(a), C8'(c) \quad (57)$$

Since the above new form is convex, the solver CVX is adopted to solve it. The specific algorithm is as Algorithm 3.

D. Convergence and Complexity Analysis

In this paper, the irregular RIS-assisted UAV network energy efficiency fairness problem in Metaverse is decomposed into three subproblems and solved by three different algorithms respectively. In the joint optimization problem of subproblem $P(B)$ and subproblem $P(C)$, the optimized solution of each subproblem is used in the algorithm of the other subproblem, which can significantly improve the optimization effect theoretically. The results obtained from the joint optimization solution of subproblem $P(B)$ and subproblem $P(C)$ are iterated to the input values of subproblem $P(A)$. The joint optimization algorithm consisting of Algorithm 2 and Algorithm 3 is iteratively optimized with Algorithm 1. Since we apply approximate optimization methods during the transformation of the three subproblems, the solution results are suboptimal solutions of the initial model. Therefore, the

optimized values of the three subproblems in this paper are defined as follows

$$\Pi^t = \begin{cases} \Pi_2^t; & P(B) \text{ is solved} \\ \Pi_{2,3}^t; & P(B) \text{ and } P(C) \text{ are solved} \\ \Pi_{1,2,3}^t; & P(A), P(B) \text{ and } P(C) \text{ are solved} \end{cases} \quad (58)$$

Thus, in the t -th iteration, we can obtain three subproblems of optimization results with $\Pi^t \leq \Pi_2^t \leq \Pi_{2,3}^t \leq \Pi_{1,2,3}^t$. It is shown that the value of the objective function tends not to decrease after each iteration. Since the initial problem has limitations such as finite UAV transmitting power and finite cross-layer interference, the original upper bound on the problem is a finite value, so that the scheme proposed in this paper can eventually converge with the following computational complexity.

The search complexity for irregular IRS topology optimization is $T_A = \text{Neigh}_S I_A$. For the subproblem of UBS power control, CVX is adopted to solve the semi-definite programming (SDP) problem. Under a SDP restraints which contains an $b \times b$ positive semi-definite matrix, the complexity is $O(\sqrt{b} \log(1/\epsilon) (ab^3 + a^2b^2 + a^3))$, where ϵ is the non-negative accuracy parameters [43]. Therefore, the complexity required to solve $P(B)$ is $T_B = O(\log(1/\epsilon)((7U_k + 2)(7U_k + 3 + (7U_k + 2)^2)))$ with $b = 1$, $a = 7U_k + 2$. Similarly, to solve the phase shift design subproblem $P(C)$, with $b = M$, $a = 4U_k + 3$, calculate the complexity of $P(C)$ as $T_C = O(\log(1/\epsilon)((4U_k + 3)(M^{3.5} + (4U_k + 3)M^{2.5} + M^{0.5}(4U_k + 3)^2)))$. Using the aforementioned analysis, the total computational complexity can be stated as follows $T_A(I_B T_B + I_C T_C)$.

IV. SIMULATION RESULTS

The key simulation settings are exhibited in Table I, where $Kr = \kappa_{dl} = \kappa_{rl}$. The central location of the MBS is within the macro cell, and K UAVs are casually distributed in the cell. The variation of the parameters in each simulation are illustrated in each figure below.

In Fig. 2, the number of UAVs is 4, and each UAV serves 20 Metaverse users. The UBS transmit power is 8 W. The simulation exhibits the convergence of EE via different cases. The method proposed shows a tendency to converge rapidly in both the regular RIS and irregular RIS scenarios. Similarly, the joint algorithm proposed under imperfect CSI and perfect CSI both converge in about 7 times, which demonstrates low complexity in Metaverse. Under the regular RIS-assisted Metaverse service, the EE performance with imperfect CSI is far less than that perfect CSI scheme. Due to perfect CSI scheme has no channel information loss, and

$$\nabla_{\tilde{\Phi}_\ell}^H f_4(\tilde{\Phi}_\ell)^{(t)} = \frac{\sum_{j=i+1}^{U_k} P_{k,j}^\ell \mathbf{A}_{k,i}^\ell (\mathbf{A}_{k,i}^\ell)^H}{\sum_{j=i+1}^{U_k} \text{tr} \left(P_{k,j}^\ell (\mathbf{A}_{k,i}^\ell)^H (\tilde{\Phi}_\ell)^{(t)} \mathbf{A}_{k,i}^\ell \right) + \sum_{\nu=1}^{U_{K+1}} \text{tr} \left(H_{k,K+1,i}^\ell P_\nu \right) + \sigma^2} \ln 2 \quad (44)$$

TABLE I
SIMULATION PARAMETERS

Parameters	Values
Macro cell radius, UAV cell radius	800 m, 200 m
Minimum distance between MBS and UAV	50 m
Number of macro Metaverse users	20
Number of UAV Metaverse users	10~20
Number of UAV cells	2~10
Height of UAV, height of RIS	100 m, 30 m
System bandwidth	1 MHz
Outage probability, α_{out}	0.05
Hover power consumption of each UAV, p_c	0.1~0.9 W
AWGN power spectral density	-174 dBm/Hz
Rician factor of RIS-UUE link, K_r	10 dB
Path-loss index of UBS-UUE and RIS-UUE link	1.6~2.4
Number of RIS elements, M	16~48
Maximum UAV transmit power P_k^{UAV}	4~12 W
Maximum transmit power of MBS	40 dBm
Minimum distance between MBS and MUE	50 m

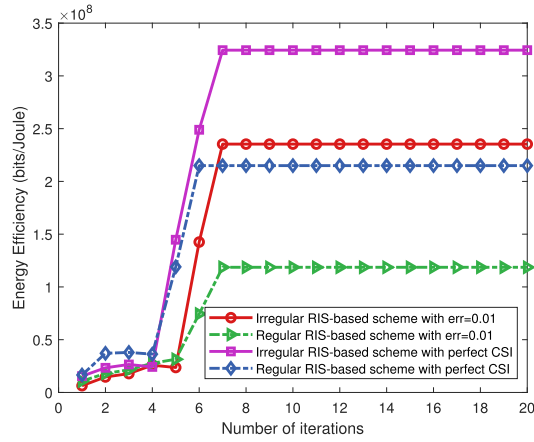


Fig. 2. The convergence of EE under different schemes.

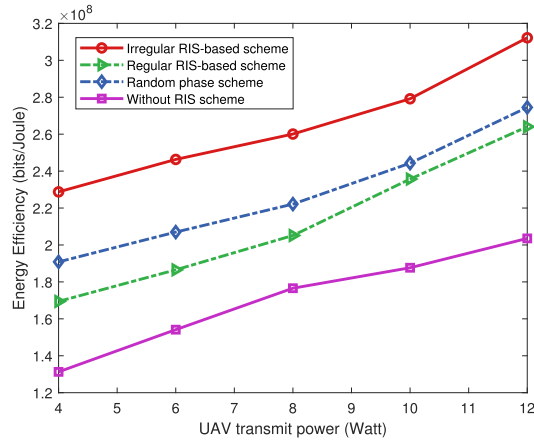


Fig. 3. EE versus different maximum transmit power of UAV.

Metaverse users can perceive precise channel information. With the same perfect channel estimation, the design of deploying irregular RISs and performing topology optimization improves the system gain by 33.7% compared to the traditional design. Due to the low cost of irregular RISs deployment and high system performance gain, deploying irregular RISs in Metaverse can play a role in economical and energy saving to a large extent.

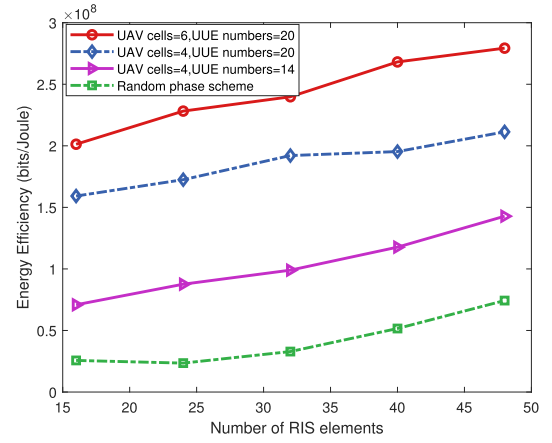


Fig. 4. EE as the number of RIS effective components varies.

Fig. 3 compares the proposed scheme with other three existing algorithms on EE as the UAV transmit power varies. The value of UBSs is 4, the value of Metaverse UUE is 14, the effective elements of RIS is 10, and the sparsity ratio is 50%. The sparsity ratio indicates the number of effective elements in the RIS as a percentage of the total number of RIS. The EE of the four schemes shows an upward trend as UAV transmit power increase. By contrast, the EE gain of irregular RIS-based scheme is 14.6% higher than Random phase scheme. This proves that the RIS phase optimization proposed has prominent benefits. In addition, compared with irregular RIS-based scheme and without RIS scheme, the EE gain of the irregular RIS design of this paper is significant. This embodies the unconventional design for RIS and topology optimization performs good in Metaverse system.

Fig. 4 exhibits the influence of various schemes on the system EE as the valid elements of RIS varies. The sparsity ratio of RIS components remains as 50%. As the effective units of RIS adding, the system gain of the four scenes shows a slow growth trend. When the number of Metaverse UUE is 20, the more the UAV cell is, the greater the gain of the whole system will be. Moreover, when the number of UAV cells is fixed at 4, the more Metaverse users each cell serves, the greater the EE earnings of the whole system. Not only that, but when RIS is deployed in the system, even if the phase is randomly assigned to the RIS components, the EE gains still increase slowly as the components of RIS enlarge. This demonstrates that even with large scale RIS deployments in Metaverse, the low complexity RIS controller design can guarantee high system gain.

Fig. 5 illustrates the rise in EE gain as the count of UAV cells grows for four schemes. The transmission power of UBS is 8 W, and each cell has 20 UUE. In contrast, the EE gain under perfect CSI scheme is markedly excelled to the EE gain under imperfect CSI scheme as the number of UAV cells increases. Note that in both regular RIS-based scheme and random phase scheme, the system gain is notably better than that the case without RIS. Moreover, the optimization of phase shift brings much more system gain than the design of unconventional RIS element topology design. This simulation effectively proves that the combination of these two designs

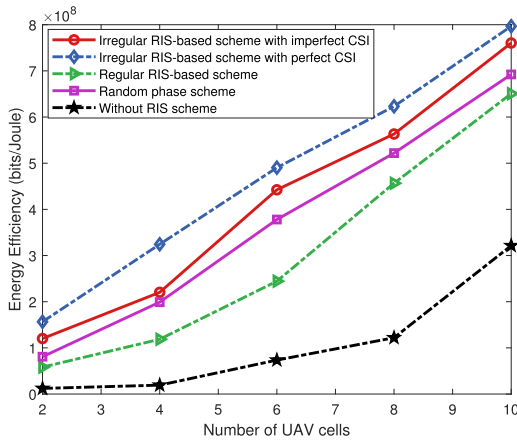


Fig. 5. EE as the number of UAV cells varies.

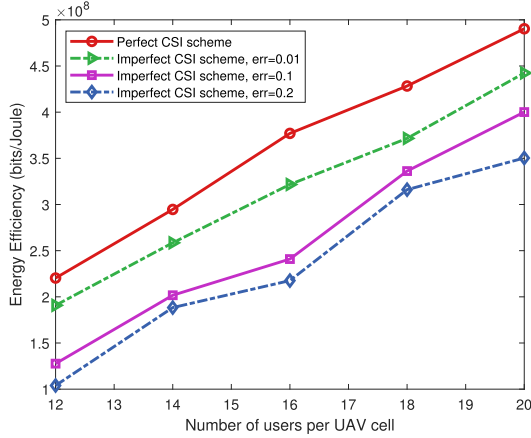


Fig. 6. EE versus the number of Metaverse UUE under different estimated error.

can double the benefits of RIS-assisted Metaverse service while maintaining low complexity, and the proposed scheme is more suitable for dense Metaverse scenes with UAV cells.

Fig. 6 evaluates EE gains versus the number of Metaverse UUE under different estimated error. No matter how large the estimation error of the cascaded CSI is, the system EE always shows a trend of steady increase as the number of UUE grows in each UAV cell. Besides, the bigger the estimation error coefficient, the less the EE gain of the system in Metaverse. When the estimation error increases, the terminal has poor capability to perceive the cascade channel information. Due to the maximum channel capacity is difficult to accurately calculate, the instantaneous rate may exceed the maximum capacity, which causes the outage probability to increase.

Fig. 7 displays the trend of the system EE as path-loss index of RIS-UUE link increases. The hovering power consumption and transmit power of UAV is 0.1 W and 8 W, respectively. Obviously, as the path-loss index increases, the system EE in the four cases shows a rapid decline trend. The reflecting signal becomes weaker owing to the growing path-loss of the RIS-UUE link. In addition, the larger the Rician factor, the greater the EE gain of the Metaverse service system. Either in irregular RIS-based case or regular RIS-based case, the higher Rician factor K_r , the better the system EE gain.

Fig. 8 presents the impact of EE gains as UAV hover power consumption change. In Fig. 8, the network capability drops

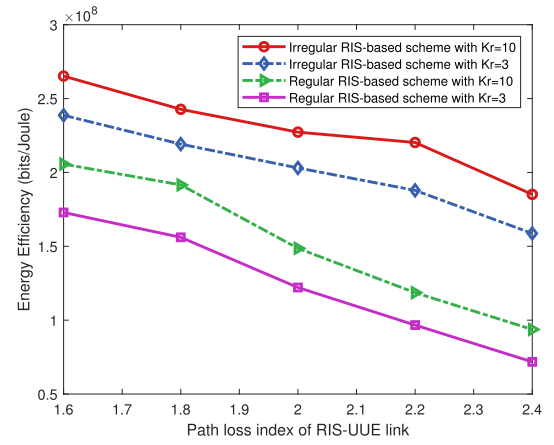


Fig. 7. EE versus path-loss index of RIS-UUE link.

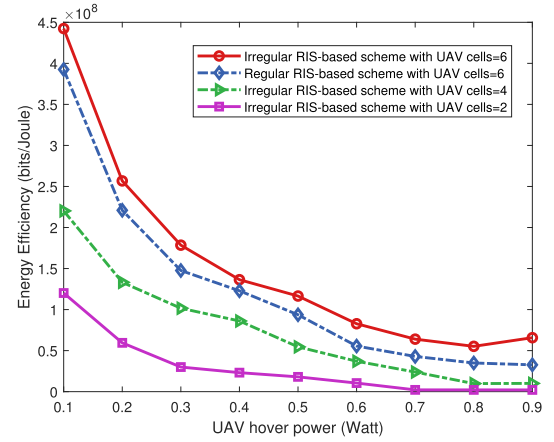


Fig. 8. EE versus UAV hover power consumption.

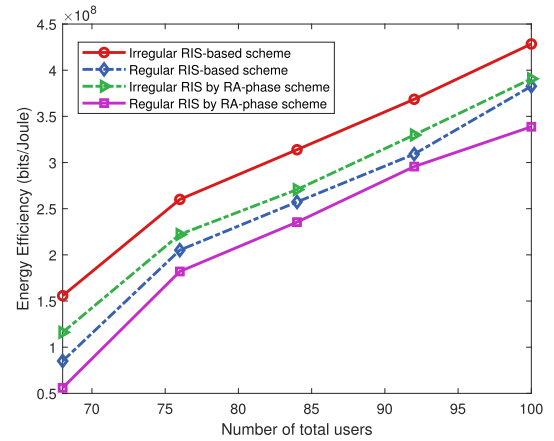


Fig. 9. EE versus the number of total Metaverse users under different scenarios.

rapidly as the UAV hover power increases. This is due to an increase in hover power results in an enhancement in total power consumption. As a result, the proportion of whole rate to total power loss is reduced. With the same number of UAV cells, irregular RIS deployment decreases rapidly as the UAV hovering power rises. Nevertheless, the scene of irregular RIS still outperforms the regular RIS schemes. Finally, as the hovering power of UAVs elevates, the UAV cells' EE gain of the whole number of UAVs gradually reduces.

In Fig. 9, we explore the EE gain of various cases as the total value of Metaverse users increase. The value of UAV cells is 4. Fig. 9 illustrates as the total Metaverse users value elevates, the EE gains under irregular RIS deployments are always better than those under traditional RIS deployments. This is due to the RIS topology optimization algorithm designed in this paper. Besides, compare irregular RIS-based scheme with irregular RIS by random phase (RA-phase) scheme, the phase shift optimization design scheme of the RIS elements designed in this paper also produces a stable EE gain return. Fig. 9 demonstrates that the irregular RIS deployment designed in this paper is suitable for dense Metaverse user areas, which is in line with practical requirements of Metaverse service.

V. CONCLUSION

Considering the constraint of rate outage probability and cross-layer interference, this paper focuses on RIS topology matrix design, power control and low complexity phase shift design of irregular RIS to obtain the maximum EE while ensuring fairness. Since the primary problem is a non-convex function in fractional form, we transform it into a subtraction form to simplify the solution. Based on tabu search algorithm, an effective topology optimization scheme for irregular RIS is proposed. Then, the outage probability constraint is transformed into convex optimization constraint by Bernstein type inequality. In addition, employing the penalty method to improve the phase shift of RISs, and significant performance gain is obtained. Finally, the validity of the proposed co-design is certified by plentiful simulation experiments. Meanwhile, the influence of low complexity phase-shift and topology approach of irregular RIS on system EE in Metaverse is not negligible.

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